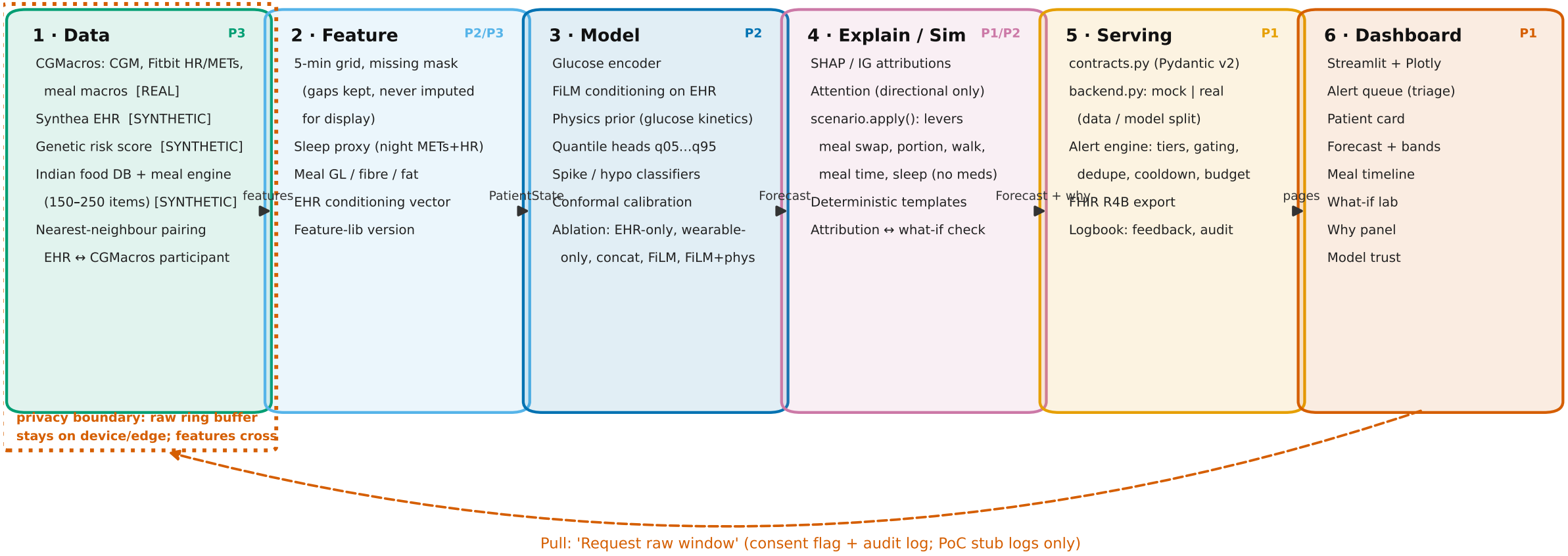
MadhuTwin: architecture (7 layers)

Research PoC on synthetic/open data. Owners: P1 frontend/integration · P2 model · P3 data. Dashed = PoC stub or future scope.



7 · Governance (cross-cutting)

Provenance label on every number: REAL / SYNTHETIC / MOCK / MODEL ESTIMATE · synthetic-only data rule · pseudonymous IDs (P-###), no names or addresses

Model + feature-lib version on every alert and FHIR resource · clinician feedback stored for a future federated loop · consent flag + audit log for Pull

Model-trust page: calibration, subgroups, Clarke grid, ablation, honest limits · what-if labelled 'model estimate, not causal, not medical advice'

FUTURE SCOPE (not built): federated learning, ABDM integration, MQTT edge transport in production